Why Soap Films are smarter than Calculus

A dive into the history of Plateau's Problem

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Abstract

Dip a wire frame into soapy water, and it instantly solves a complex area-minimization problem that stumped mathematicians for over a century. Why did it take us so long to catch up? Because classical calculus of variations assumes surfaces are well-behaved, smooth graphs. Unfortunately, soap films are topological rebels: they branch, they intersect, and they form singularities that break simple functional minimization.

In these notes, we will trace the history of Plateau's Problem, moving from the limitations of classical analysis to the modern advances of Geometric Measure Theory. We will discuss why we had to redefine what a surface is, introducing concepts like Currents and Varifolds to handle the orientation and multiplicity issues that arise in physical films. Finally, we will see how the framework of (M, ε, δ)-minimal sets provides the rigorous structure necessary to prove Plateau's empirical laws.

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1 What is a Soap Film?

To understand the mathematical complexity of soap films, we must first appreciate the physical ingenuity of the material itself. A soap film is not merely a static geometric surface $S \subset \mathbb{R}^3$; it is a dynamic, fluid system that actively works to maintain its existence against gravity and evaporation.

Physically, a stable soap film is a solution composed of three essential ingredients:

- **distilled water**, the bulk medium;
- **surfactant** (most commonly soap), the active agent that lowers surface tension;
- **stabilizer** (most commonly glycerin), a viscous agent that prolongs the film's life.

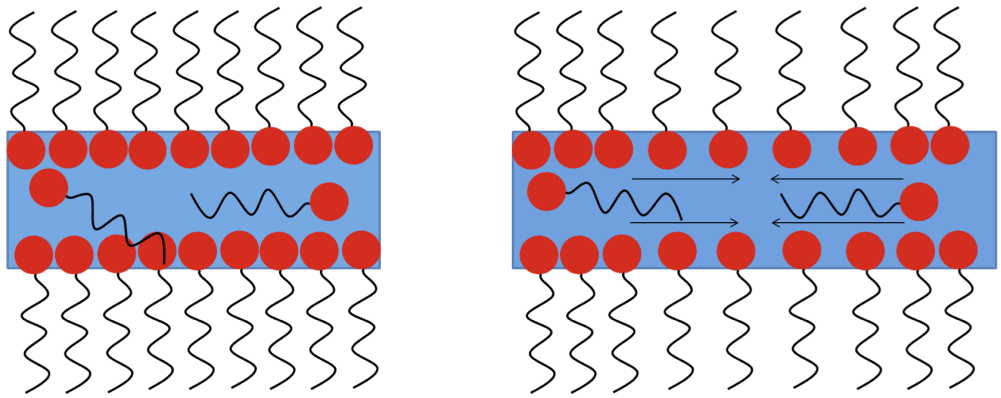

Figure 1: Cross-section of a soap film and surfactant distribution (left) with Marangoni effect (right).

The magic lies in the molecular structure of the surfactant. These molecules are **amphiphilic**, meaning that they possess a dual nature consisting of:

- a **hydrophilic** head, stable when interacting with water;
- a **hydrophobic** tail, stable when isolated from water.

When dissolved in water, these molecules arrange themselves to minimize energy. In a thin film, they form a "sandwich" structure where the aqueous core sits in the middle, while the surfactant molecules line the two air-water interfaces. Crucially, they orient themselves with their hydrophilic heads submerged in the water core and their hydrophobic tails protruding outward into the air (see Fig. 1 on the left).

It is well known that pure water cannot form stable films; its high surface tension causes any thin sheet to rupture instantly. Soap films instead survive due to a phenomenon known as the **Marangoni effect** (or Gibbs-Marangoni elasticity).

The mechanism is a brilliant feedback loop (see Fig. 1 on the right):

1. imagine that a fluctuation (air current or gravity) stretches a patch of the film, making it thinner;
2. as the area expands, the surface concentration of surfactant molecules in that patch decreases (the average distance between molecules increases);
3. since surface tension decreases as surfactant concentration increases, the local surface tension *increases* in the thinned region;

4. this gradient in tension creates an elastic restoring force (tangential to the surface) that pulls liquid from the surrounding (thicker) regions back into the thinned spot, stabilizing it.

The addition of glycerin, or any other stabilizer, increases the viscosity of the water core, slowing down drainage due to gravity, and enhances the Marangoni effect. Additionally, glycerin modifies the refractive index, enhancing the thin-film interference that produces the psychedelic colours we associate with soap bubbles.

This physical drive to minimize surface tension translates directly into mathematics. The total energy of the film is proportional to the number of surfactant molecules exposed at the interface, which is directly proportional to the surface area.

Therefore, the film naturally seeks an equilibrium state that minimizes the surface area:

$$E(S) = 2\gamma \cdot \text{Area}(S), \tag{1}$$

where γ is the surface tension coefficient. (The factor of 2 arises because the film comprises two liquid-air interfaces).

Before we attack this with calculus, it is worth noting: the liquid solves this non-linear PDE instantly, purely through molecular interactions, often finding solutions that are topologically complex or singular—solutions that classical calculus struggles to even define. As we will see, the main difficulties lie in understanding what is the correct meaning of *surface*.

In the absence of a boundary, this minimization results in a sphere (a bubble). This is relatively simple in isolation, but the problem of understanding the geometry of a network of bubbles, such as a foam, is far from trivial.

When we introduce a boundary, like a twisted wire or a complex frame, the film solves a constrained minimization problem. It must find the surface of least area spanning that boundary, a problem that naturally leads to the condition of zero mean curvature.

This condition arises from another perspective as well: the pressure difference across the film. The **Young-Laplace equation** states that the pressure difference between the two sides of the film is proportional to the mean curvature:

$$\Delta p = 4\gamma \cdot H, \tag{2}$$

where H is the mean curvature.

For a film spanning a wire (unlike a closed bubble), the pressure is atmospheric on both sides, so $\Delta p = 0$. This forces the condition $H = 0$. Therefore, a minimal requirement for a soap film model to be physical is that it predicts $H = 0$ on the regular parts of the surface.

2 Real-life examples and Plateau's laws

In the presence of a prescribed boundary Γ, we can state the problem formally as follows:

Setup: Let $\Gamma \subset \mathbb{R}^3$ be a closed Jordan curve. Find a surface S such that:

1. $\partial S = \Gamma$;
2. $\text{Area}(S) \leq \text{Area}(S')$, for all surfaces S' with $\partial S' = \Gamma$.

It seems simple enough, right? Let's see what happens in the case of a simple geometry, like a flat disk. As you can see from Fig. 2 (left), the solution is what we would expect: a planar disk.

Well, let's now bend the wire frame slightly. As we can see from Fig. 2 (centre), we get a surface resembling a saddle. This shape shape has negative Gaussian curvature, allowing it to maintain zero mean curvature everywhere.

Let's make this harder: take a circular frame and bend it so that it becomes highly non-planar. Now, it is quite difficult to guess what shape the film will take. Indeed, from Fig. 2 (right), we can see a more complex geometry emerging.

Figure 2: Soap films for disk-like frames: planar (left), saddle (centre), and highly non-planar surface (right).

We can generate more exotic examples, especially if we allow the frame to be a general 1-dimensional object in $\mathbb{R}^3$, where the resulting film is even wilder:

1. **catenoid**: The minimal surface spanning two parallel coaxial rings. It is the only minimal surface of revolution (see Fig. 3, left).
2. **helicoid**: A ruled surface resembling a spiral staircase, extending to infinity (locally minimal) (see Fig. 3, center).
3. **Möbius strip**: A non-orientable surface bounded by a single curve (e.g., a wire twisted twice) (see Fig. 3, right).
4. **singular structures**: When dipping a wire frame in the shape of a tetrahedron or a cube, we observe internal structures where surfaces meet. These involve singularities (edges and vertices) that classical calculus struggles to define (see Fig. 4, left-center).
5. **Steiner-like structures**: For certain boundaries, the film forms Y-shaped branching junctions to minimize area, analogous to the Steiner Tree problem in graph theory (see Fig. 4, right).

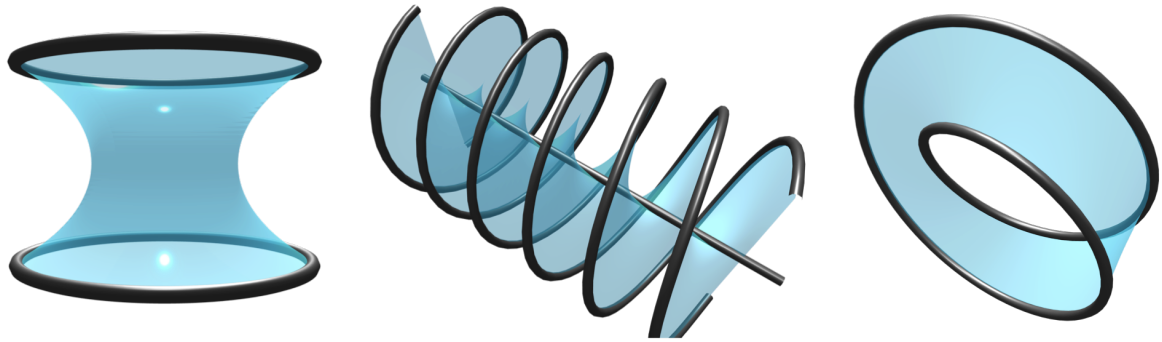

Figure 3: Soap films for catenoid (left), helicoid (centre), Möbius strip (right).

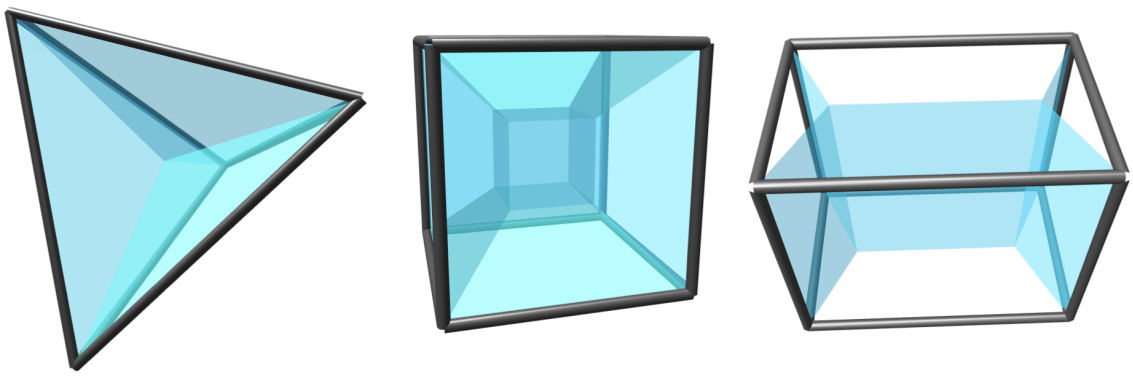

Figure 4: Soap films for tetrahedron (left), cube (centre), Steiner structure (right).

Looking at all this topological mess, you might get discouraged. However, in 1873 Joseph Plateau did not. He attempted to create a complete empirical theory of soap films. By dipping thousands of various wire frames into soap solutions, he catalogued the resulting forms and inferred universal physical laws.

Plateau's observations provided the *ground truth* that mathematicians took over a century to prove.

Plateau's Laws (1873):

1. **Local Minimality**: Away from singularities, soap films are smooth surfaces that locally minimize area (zero mean curvature).

2. **Triple Junctions**: Films meet in groups of three along smooth curves (called *Plateau borders*). The tangent planes meet at dihedral angles of $\arccos(-1/2) = 120°$.

3. **Tetrahedral Vertices**: These Plateau borders meet in groups of four at a single vertex. The angle formed by the borders is the tetrahedral angle, $\arccos(-1/3) \approx 109.47°$.

These laws imply that the singularities can be of only two kinds, which we will refer to as **Triple Junctions** (type $\mathbb{Y}$) and **Tetrahedral Points** (type $\mathbb{T}$). Far from these points, the film is locally a plane (type $\mathbb{P}$).

Any successful theory of soap films has to accurately predict these laws, and as we will see, achieving this rigour required the development of entirely new fields of mathematics.

3 Graphs

Even before Plateau's laws were formulated, mathematicians knew that to find the surface spanning a given boundary, they had to minimize surface area (recall (1)). The problem was first proposed by Leonhard Euler and Joseph-Louis Lagrange, and solved by Lagrange in 1760 by restricting the search to surfaces that are graphs of functions.

Assume S is the graph of a function $u\colon \Omega \subset \mathbb{R}^2 \to \mathbb{R}$, and assume that the prescribed boundary Γ is the graph of a function $g\colon \partial\Omega \to \mathbb{R}$. The boundary condition is simply $u|_{\partial\Omega} = g$.

The surface is parametrized by

$$S := \left\{ r(x,y) = \big(x, y, u(x,y)\big) : (x,y) \in \Omega \right\}.$$

The tangent vectors on the surface are

$$r_x(x,y) = \big(1, 0, u_x(x,y)\big), \qquad r_y(x,y) = \big(0, 1, u_y(x,y)\big).$$

The area element is given by the magnitude of the cross product of these tangent vectors. Thus, the area functional is:

$$\begin{aligned} A(u) &= \int_\Omega |r_x \times r_y| \,\mathrm{d}x\,\mathrm{d}y \\ &= \int_\Omega |(-u_x, -u_y, 1)| \,\mathrm{d}x\,\mathrm{d}y \\ &= \int_\Omega \sqrt{1 + |\nabla u|^2} \,\mathrm{d}x\,\mathrm{d}y. \end{aligned} \tag{3}$$

We seek a function u, constrained at the boundary, that minimizes this functional. Assuming such a minimum exists and is sufficiently smooth, we compute the Euler-Lagrange equation.

Consider a variation $v \in C_c^\infty(\Omega)$. For any $t \in \mathbb{R}$, the function $u + tv$ represents a competing surface satisfying the same boundary conditions. Define $I(t) = A(u + tv)$. If u is minimal, then $I'(0) = 0$.

$$\begin{aligned} I'(0) &= \frac{d}{dt}\bigg|_{t=0} \int_\Omega \sqrt{1 + |\nabla(u+tv)|^2} \,\mathrm{d}x\,\mathrm{d}y \\ &= \int_\Omega \frac{(\nabla u + t\nabla v)\cdot \nabla v}{\sqrt{1 + |\nabla u + t\nabla v|^2}}\bigg|_{t=0} \mathrm{d}x\,\mathrm{d}y \\ &= \int_\Omega \frac{\nabla u \cdot \nabla v}{\sqrt{1 + |\nabla u|^2}} \,\mathrm{d}x\,\mathrm{d}y. \end{aligned}$$

Integrating by parts (Green's formula) and noting that v vanishes on $\partial\Omega$, we obtain:

$$-\int_\Omega \operatorname{div}\left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}}\right) v \,\mathrm{d}x\,\mathrm{d}y = 0.$$

Since this holds for all test functions v, we derive the **Minimal Surface Equation**:

$$\mathcal{M}(u) := \operatorname{div}\left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}}\right) = 0. \tag{4}$$

Thus, a minimizer must satisfy the Dirichlet problem for this non-linear elliptic PDE:

$$\begin{cases} \mathcal{M}(u) = 0 & (x,y) \in \Omega, \\ u = g & (x,y) \in \partial\Omega. \end{cases}$$

While this PDE is readily solvable numerically, its geometric meaning is profound. We now prove that $\mathcal{M}(u) = 0$ is equivalent to the condition that the mean curvature H vanishes everywhere.

By expanding the divergence term, we explicitly write the operator as:

$$\mathcal{M}(u) = \frac{(1+u_y^2)u_{xx} - 2u_x u_y u_{xy} + (1+u_x^2)u_{yy}}{(1+|\nabla u|^2)^{\frac{3}{2}}}. \tag{5}$$

From Differential Geometry, the mean curvature H is given by the trace of the shape operator (the product of the Second and the inverse of the First fundamental forms):

$$2H = \operatorname{trace}\left(\mathrm{II} \cdot \mathrm{I}^{-1}\right),$$

where

$$\mathrm{I} = \begin{pmatrix} E & F \\ F & G \end{pmatrix}, \qquad \mathrm{II} = \begin{pmatrix} L & M \\ M & N \end{pmatrix}.$$

The formula becomes:

$$2H = \frac{EN + GL - 2FM}{EG - F^2}.$$

Let $W = \sqrt{1+|\nabla u|^2}$. We compute the coefficients of the first fundamental form:

$$E = r_x \cdot r_x = 1 + u_x^2, \qquad F = r_x \cdot r_y = u_x u_y, \qquad G = r_y \cdot r_y = 1 + u_y^2.$$

Note that the determinant is $EG - F^2 = 1 + u_x^2 + u_y^2 = W^2$. For the second fundamental form, we require the unit normal vector ν:

$$\nu := \frac{r_x \times r_y}{|r_x \times r_y|} = \frac{(-u_x, -u_y, 1)}{W}.$$

The coefficients are projections of the second derivatives onto the normal:

$$L = r_{xx} \cdot \nu = \frac{u_{xx}}{W}, \quad M = r_{xy} \cdot \nu = \frac{u_{xy}}{W}, \quad N = r_{yy} \cdot \nu = \frac{u_{yy}}{W}.$$

Substituting these into the formula for H:

$$2H = \frac{(1+u_x^2)\frac{u_{yy}}{W} + (1+u_y^2)\frac{u_{xx}}{W} - 2(u_x u_y)\frac{u_{xy}}{W}}{W^2} = \mathcal{M}(u).$$

Therefore, area minimization implies $H = 0$ everywhere.

This result confirms Plateau's first law (zero mean curvature) but highlights the limitations of the graph approach. The equation $H = 0$ holds perfectly for the "flat disk" and "slightly bent" examples in Fig. 2, which are representable as graphs. This principle is also the basis for Mean Curvature Flow algorithms, where a surface evolves according to $\frac{\partial S}{\partial t} = \mathbf{H}$, effectively performing gradient descent on the area functional.

However, the graph formulation has severe drawbacks:

- **No vertical tangents**: the gradient $|\nabla u|$ cannot be infinite. This excludes surfaces like the Catenoid (Fig. 3), which has vertical tangents at its waist.
- **No self-intersections**: graphs must be single-valued, excluding complex topologies.
- **No singularities**: it cannot represent the Y-junctions or tetrahedral points essential to Plateau's laws.

4 Parametric Surfaces

To overcome the limitations of the graph approach (specifically the inability to handle vertical tangents and complex topology), Jesse Douglas and Tibor Radó independently found a solution in 1930–1931 by viewing soap films as **parametrized surfaces**.

Their methods were quite different. Radó's work built on the previous results of René Garnier and held for rectifiable simple closed curves. Douglas used completely new ideas, producing a result valid for *arbitrary* simple closed curves. Both relied on setting up minimization problems: Douglas minimized the functional now known as the Douglas Integral, while Radó minimized the Dirichlet Energy. Douglas was awarded the first Fields Medal in 1936 for this work. The approach presented here follows Radó's energy method, which is more standard in modern PDEs.

Let $u\colon D \subset \mathbb{R}^2 \to \mathbb{R}^3$ be the map representing our surface, where D is the unit disk, satisfying the boundary constraint $u(\partial D) = \Gamma$. The area functional is:

$$A(u) = \int_D |u_x \times u_y| \,\mathrm{d}x\,\mathrm{d}y.$$

There is a significant difficulty with minimizing this functional that was not present in the graph approach.

- *Why wasn't it a problem before?* In the graph approach, the parametrization is fixed to the specific form $(x, y) \mapsto (x, y, h(x, y))$. We cannot reparametrize the domain without changing the problem setup.
- *Why is it a problem now?* The area functional is **invariant under diffeomorphism** (reparametrization) of the domain D.

To see this, let $\varphi\colon D \to D$ be a diffeomorphism. The area element is given by the square root of the determinant of the induced metric (First Fundamental Form) $g_{ij} = \partial_i u \cdot \partial_j u$. Let g_u denote the matrix of this metric. Then:

$$|u_x \times u_y| = \sqrt{\det g_u}.$$

Define a reparametrized surface $\widetilde{u} := u \circ \varphi$. By the chain rule, the new metric is

$$g_{\widetilde{u}} = (\mathrm{D}\varphi)^{\mathrm{T}}(g_u \circ \varphi)\mathrm{D}\varphi.$$

Taking the determinant:

$$\sqrt{\det g_{\widetilde{u}}} = |\det \mathrm{D}\varphi|\sqrt{(\det g_u) \circ \varphi}.$$

Using the change of variables formula for integration:

$$\begin{aligned} A(u \circ \varphi) &= \int_D \sqrt{\det g_{\widetilde{u}}}\,\mathrm{d}x\,\mathrm{d}y \\ &= \int_D |\det \mathrm{D}\varphi|\sqrt{(\det g_u) \circ \varphi}\,\mathrm{d}x\,\mathrm{d}y \\ &= \int_{\varphi(D)} \sqrt{\det g_u}\,\mathrm{d}x\,\mathrm{d}y = A(u). \end{aligned}$$

This invariance causes a **loss of compactness**. Given a map $u \in W^{1,2}(D; \mathbb{R}^3)$, one can construct a sequence of diffeomorphisms $\{\varphi_n\}_{n\in\mathbb{N}}$ that "bunch up" the derivatives such that:

$$A(u \circ \varphi_n) = A(u), \qquad \text{but} \quad \|u \circ \varphi_n\|_{W^{1,2}} \to \infty.$$

We lose coercivity, meaning we cannot apply the Direct Method of Calculus of Variations to $A(u)$.

Radó's solution relies on the elementary algebraic inequality:

$$|u_x \times u_y| \leq \frac{1}{2}(|u_x|^2 + |u_y|^2) = \frac{1}{2}|\nabla u|^2,$$

where equality holds if and only if the map is **conformal** ($|u_x| = |u_y|$ and $u_x \cdot u_y = 0$). This gives us a relation between Area and Dirichlet Energy:

$$A(u) \leq E(u) := \frac{1}{2}\int_D |\nabla u|^2 \,\mathrm{d}x\,\mathrm{d}y.$$

Radó's insight was to minimize $E(u)$ instead. The question is: if we find a minimizer u^* for $E(u)$, is it also a minimizer for $A(u)$? This happens precisely if $A(u^*) = E(u^*)$, which requires u^* to be conformal.

Theorem 4.1 (Douglas-Radó, 1931)**.** *Let Γ be a continuous simple closed Jordan curve in $\mathbb{R}^3$. Then there exists a map $u \in C(\overline{D};\mathbb{R}^3) \cap C^2(D;\mathbb{R}^3)$ mapping ∂D monotonically onto Γ, such that u is:*

1. ***Harmonic:*** *$\Delta u = 0$ in D;*
2. ***Conformal:*** *$|u_x| = |u_y|$ and $u_x \cdot u_y = 0$;*
3. ***Area Minimizing:*** *u minimizes $A(v)$ among all disk-type surfaces spanning Γ.*

This result was later extended to surfaces of higher topological complexity (planar domains with holes, handling genus 0 and connectivity $k > 1$), such as the catenoid.

Douglas's approach differed by exploiting the property that the energy of a harmonic function is determined entirely by its boundary values. If $\gamma\colon \partial D \to \mathbb{R}^3$ is the boundary parameterization, the energy of the harmonic extension is given by the **Douglas integral**:

$$\mathcal{A}(\gamma) := \frac{1}{4\pi}\int_0^{2\pi}\int_0^{2\pi} \frac{|\gamma(t) - \gamma(s)|^2}{\sin^2\left(\frac{t-s}{2}\right)} \,\mathrm{d}t\,\mathrm{d}s.$$

Minimizing this integral over monotonic parametrizations yields the boundary values of the minimal surface.

Crucially, the Douglas-Radó approach also predicts zero mean curvature. Recall the formula for the Laplace-Beltrami operator Δ_{I} applied to the position vector u:

$$\Delta_{\mathrm{I}} u = 2H\nu,$$

where ν is the orientation vector. If u is conformal, the induced metric is $\mathrm{I}_{ij} = \lambda^2\delta_{ij}$. The operator scales as:

$$\Delta_{\mathrm{I}} u = \frac{1}{\lambda^2}\Delta u.$$

Thus, we have the relation:

$$\Delta u = 2\lambda^2 H\nu.$$

Since the minimizer of the Dirichlet Energy is harmonic ($\Delta u = 0$), it follows that $H = 0$ everywhere (where $\lambda \neq 0$).

Drawbacks: While this method solves the issues of vertical tangents and existence, it remains topologically limited:

- **Fixed topology**: it finds a disk-type surface. It cannot automatically detect if the physical soap film would prefer to be a Möbius strip or a higher-genus surface.
- **No topological singularities**: It produces an immersion (possibly with branch points where $\nabla u = 0$), but it cannot model the **Triple Junctions** (Y-shapes) observed in Plateau's laws. The model forces the surface to be a smooth manifold everywhere, contradicting physical reality at the junctions.
- **Orientation**: it requires orientation, therefore it cannot model non-orientable shapes like the Möbius strip.

5 Geometric Measure Theory

To handle the singularities observed by Plateau and to free ourselves from fixed topological constraints, we need a generalized notion of a "surface" that retains good minimization properties. We need an object that behaves like a measure (to measure area) but retains vector properties (to handle orientation and Stokes' theorem).

In 1960, Herbert Federer and Wendell Fleming revolutionized the field by defining the class of **Integral Currents**. Their idea was to view surfaces not as maps from a domain, but as functional objects acting on differential forms, essentially generalizing the concept of distributions to geometry.

Definition 5.1. *Let $\mathcal{D}^k(M)$ be the space of smooth, compactly supported differential k-forms on a smooth oriented n-dimensional manifold M. A **k-current** T is a continuous linear functional on $\mathcal{D}^k(M)$ (in the sense of distributions). The space of k-currents is denoted by $\mathcal{D}_k(M)$.*

Just as distributions have a support, so do currents.

Definition 5.2 (Support)**.** *The **support** of a current T, denoted $\operatorname{spt}(T)$, is the smallest closed set $C \subset M$ such that $T(\omega) = 0$ for all forms ω with support disjoint from C.*

The power of this definition lies in how easily it generalizes classical calculus operations. For instance, the boundary of a surface is defined purely by duality using Stokes' Theorem. If T is a k-current, its boundary ∂T is the $(k-1)$-current defined by:

$$(\partial T)(\omega) \coloneqq T(\mathrm{d}\omega) \qquad \forall \omega \in \mathcal{D}^{k-1}(M).$$

To talk about Plateau's problem, we need to measure "area." We define the mass of a current via the dual norm:

$$\mathbb{M}(T) \coloneqq \sup\{T(\omega) : \omega \in \mathcal{D}^k(M), \|\omega\|_\infty \leq 1\}.$$

Here, $\|\omega\|_\infty$ is the co-mass norm (the supremum of the form's value on unit simple k-vectors). If T represents integration over a smooth surface, $\mathbb{M}(T)$ is exactly the surface area.

The space of all currents is too large (it contains "fuzzy" distributional objects). We want to restrict ourselves to currents that look like geometric surfaces.

Consider a countably k-rectifiable set $S \subset M$ (a set that is contained in a countable union of C^1 submanifolds up to a set of $\mathcal{H}^k$-measure zero). We can equip S with:

1. An orientation $\xi(x)$, which is a simple unit k-vector field tangent to S (defined $\mathcal{H}^k$-a.e.);
2. A multiplicity function $\theta(x) \in \mathbb{N}$, representing the number of layers (folds) at x.

This triplet (S, ξ, θ) defines a **Rectifiable Current** via integration:

$$T(\omega) = \int_S \langle \omega(x), \xi(x) \rangle \theta(x) \, \mathrm{d}\mathcal{H}^k(x). \tag{6}$$

In this case, the mass is simply the weighted area: $\mathbb{M}(T) = \int_S \theta \, \mathrm{d}\mathcal{H}^k$.

However, for the calculus of variations, we need compactness. A sequence of rectifiable currents might converge to something with a "wild" boundary. To fix this, Federer and Fleming defined the class of **Integral Currents**, denoted $\mathbf{I}_k(M)$:

Definition 5.3. *A current T is an **Integral k-Current** if both T and its boundary ∂T are rectifiable currents (with integer multiplicities).*

This condition ensures that we don't just have layers of film (T), but the rim of the film (∂T) is also well-behaved.

The cornerstone of Geometric Measure Theory, and the reason it succeeded where classical calculus failed, is the compactness theorem.

Theorem 5.1 (Federer-Fleming Compactness, 1960)**.** *Let $\{T_j\}_{j\in\mathbb{N}}$ be a sequence of integral currents in a compact set K such that:*

$$\sup_j \left(\mathbb{M}(T_j) + \mathbb{M}(\partial T_j)\right) < \infty.$$

*Then, there exists a subsequence converging in the **flat norm** topology to an integral current T.*

The **flat norm** measures the geometric distance between currents:

$$\mathcal{F}(T) := \inf\{\mathbb{M}(A) + \mathbb{M}(B) : T = A + \partial B\}.$$

Intuitively, if T_1 and T_2 are close in flat norm ($T_1 - T_2 = A + \partial B$), it means we can deform T_1 into T_2 by adding a small patch A (error in volume/filling) and a small boundary B (error in area/smoothing).

Because the mass functional is lower semi-continuous with respect to this topology, the Direct Method of Calculus of Variations applies immediately.

Corollary 5.1. *For any integral $(k-1)$-current boundary B, there exists an integral k-current T such that $\partial T = B$ and T minimizes mass.*

We found a minimizer! But is it a soap film? Does it look like a smooth surface? The regularity theory for integral currents is deep and surprising.

Theorem 5.2 (De Giorgi-Federer-Simons)**.** *Let T be a mass-minimizing integral $(n-1)$-current in $\mathbb{R}^n$.*

- *If $n \leq 7$, the support $\operatorname{spt}(T)$ is a smooth embedded hypersurface (analytic in the interior).*
- *If $n \geq 8$, singularities may exist. The singular set has Hausdorff dimension at most $n-8$.*

This theorem highlights a distinct difference between low and high dimensions. The famous counterexample in $\mathbb{R}^8$ is the **Simons Cone** (independently discovered by Bombieri, De Giorgi, and Giusti):

$$C = \{x \in \mathbb{R}^8 : x_1^2 + x_2^2 + x_3^2 + x_4^2 = x_5^2 + x_6^2 + x_7^2 + x_8^2\}.$$

This is a cone over $\mathbb{S}^3 \times \mathbb{S}^3$, has zero mean curvature and is a global minimizer, yet it has a singularity at the origin (where the two spheres meet).

Even though we work with rough objects (measures), the minimization principle still implies zero mean curvature in a weak sense. Recall the first variation formula. If T minimizes mass, then for any smooth vector field X with compact support (generating a flow ϕ_t):

$$\frac{d}{dt}\Big|_{t=0} \mathbb{M}((\phi_t)_\# T) = 0.$$

Using the area formula, this derivative is given by $\int \operatorname{div}_T X \, d\|T\|$. For a smooth surface, divergence theorem tells us $\int \operatorname{div}_S X = -\int \langle X, \vec{H}\rangle$. Thus, for a minimizer, the distributional mean curvature is zero:

$$\int \operatorname{div}_T X \, \mathrm{d}\|T\| = 0 \quad \forall X.$$

This confirms that integral currents recover the physical $H = 0$ condition.

However, for $n = 3$, the De Giorgi-Federer-Simons theorem is a double-edged sword. It proves that a minimizer in $\mathbb{R}^3$ is a **smooth embedded manifold everywhere**.

- **The Problem:** Real soap films have singularities! Plateau's laws describe Y-junctions (3 sheets meeting) and T-junctions (4 edges meeting).
- **The Contradiction:** The regularity theorem says these singularities *cannot exist* in a mass-minimizing integral current. An integral current sees a Y-junction as "three sheets," but the homological boundary condition forces cancellations or separates them in a way that minimizes area by removing the junction.

Thus, while Integral Currents are perfect for finding the minimal area, they are paradoxically *too regular* to model the complex branching structures of physical soap films. Furthermore, like the parametric approach, they require orientation, failing to model Möbius strips.

6 Varifolds

As we saw in the previous section, Integral Currents were a major step forward, but their reliance on *orientation* makes them unsuitable for modeling non-orientable soap films like the Möbius strip. Furthermore, they enforce cancellations that physically shouldn't happen.

To address this, Frederick J. Almgren Jr. and William K. Allard developed the theory of **Varifolds** (a term coined by L. C. Young, combining "manifold" and "variety"). The key idea is to discard orientation entirely and work with unoriented tangent planes.

Definition 6.1. *Let* $\mathrm{Gr}(n,k)$ *be the Grassmannian manifold of unoriented* k*-dimensional subspaces of* $\mathbb{R}^n$*. A* k***-varifold*** V *is a Radon measure on the product space* $\mathbb{R}^n \times \mathrm{Gr}(n,k)$*.*

Intuitively, a varifold tells you how much "surface weight" exists at a point x with a specific tangent direction S. We define the **mass measure** $\|V\|$ on $\mathbb{R}^n$ by projecting V onto the spatial coordinate:

$$\|V\|(A) \coloneqq V(A \times \mathrm{Gr}(n,k)) \quad \text{for any Borel set } A \subset \mathbb{R}^n.$$

The total mass is $\mathbb{M}(V) = \|V\|(\mathbb{R}^n)$.

While general varifolds can be "diffuse" (like a probability cloud of tangent planes), we are interested in geometric ones. A varifold V is called k-**rectifiable** if it is supported on a countably k-rectifiable set M with integer multiplicity. Specifically:

$$V(\phi) = \int_M \phi(x, T_x M)\, \theta(x)\, \mathrm{d}\mathcal{H}^k(x) \quad \forall \phi \in C_c(\mathbb{R}^n \times \mathrm{Gr}(n,k)).$$

Here:

- M is a countably k-rectifiable set;
- $T_x M$ is the approximate tangent space at x (which exists $\mathcal{H}^k$-a.e.);
- $\theta(x) \in \mathbb{N}$ is the multiplicity density.

Thus, a rectifiable varifold is naturally associated with the measure $V = \theta \mathcal{H}^k \llcorner M \otimes \delta_{T_x M}$.

Since varifolds have no boundary operator (no orientation means no Stokes' theorem), we define "minimal" via the variational principle. Let $X \in C_c^1(\mathbb{R}^n; \mathbb{R}^n)$ be a vector field generating a flow φ_t. The first variation of V is:

$$\delta V(X) \coloneqq \frac{d}{dt}\Big|_{t=0} \mathbb{M}((\phi_t)_\# V).$$

For a rectifiable varifold, this computes the tangential divergence of the field:

$$\delta V(X) = \int_{\mathbb{R}^n} \mathrm{div}_S X(x)\, \mathrm{d}V(x,S).$$

where $\mathrm{div}_S X$ is the divergence of X projected onto the subspace S.

We say V has a **generalized mean curvature** vector H_V if the first variation can be represented as:

$$\delta V(X) = -\int \langle H_V(x), X(x) \rangle\, \mathrm{d}\|V\|(x).$$

A varifold is **stationary** if $\delta V(X) = 0$ for all compactly supported X (away from the boundary), which implies $H_V = 0$ almost everywhere.

The theory relies on a powerful compactness theorem that requires control over the mean curvature.

Theorem 6.1 (Allard's Compactness)**.** *Let $\{V_j\}$ be a sequence of integral k-varifolds in an open set U, such that for any compact $K \subset U$:*

- $\sup_{j\in\mathbb{N}} \|V_j\|(K) < \infty$ *(Mass bound);*
- $\sup_{j\in\mathbb{N}} \int_K |H_{V_j}|^p \, \mathrm{d}\|V_j\| < \infty$ *for some $p > k$ (Curvature bound).*

Then, there exists a subsequence converging in the ***weak-* topology*** *of Radon measures to a rectifiable k-varifold V with locally L^p mean curvature.*

This allows us to find minimizers. Boundary conditions are typically handled by constraint: we minimize mass among all varifolds that are "limits" of smooth surfaces spanning Γ, or by fixing the varifold outside a compact set.

But does a stationary varifold look like a soap film? Allard proved a celebrated regularity theorem:

Theorem 6.2 (Allard's Regularity Theorem)**.** *Let V be a rectifiable k-varifold with $H_V \in L^p_{loc}$ $(p > k)$. Suppose that in a ball $B_r(x_0)$, V is close to a flat plane in terms of:*

1. ***Density:*** $\Theta^k(\|V\|, x_0) \approx 1$ *(close to single multiplicity);*
2. ***Tilt-Excess:*** *The tangent planes do not oscillate too much:*

$$\frac{1}{r^k} \int_{B_r(x_0)} \|T_x M - P\|^2 \, \mathrm{d}\|V\|(x) \ll 1 \quad \textit{for some plane } P.$$

Then, in a smaller neighbourhood of x_0, V is a single-valued $C^{1,\alpha}$ graph over P.

Allard's theorem tells us that if a soap film (stationary varifold) has density 1 and is somewhat flat, it is actually a smooth minimal surface. Crucially, this implies that singularities must have density $\Theta > 1$. This is still not enough, as it does not give a classification of the admissible singularities.

- **Unstable Singularities:** The theory of stationary varifolds allows for singularities that are not area-minimizing (e.g., 4 half-planes meeting at a line), which are not observed in physical soap films.
- **Ghost Multiplicities:** The theory allows for integer multiplicities (e.g., a "double density" plane) which are physically indistinguishable from a single thick film but mathematically distinct.
- **No Boundary Operator:** Enforcing $\partial V = \Gamma$ is technically difficult compared to currents.
- **Hard Regularity:** Proving that a minimizer doesn't have "wild" singularities requires extremely difficult dimension-reduction arguments (Almgren's Big Regularity Paper).

7 (M, ε, δ)-Minimal Sets

To finally prove Plateau's laws, we need to shift perspective. The paradigm up to now was that soap films are stable because they are global minimizers (calculus of variations). What if we turn this on its head and return to a more physical model? Soap films exist because they are **stable** with respect to local perturbations.

Idea: A set E is a soap film if, when we locally deform it, the area does not decrease significantly.

Definition 7.1 (Deformation)**.** *Let $U \subset \mathbb{R}^n$ be an open set. A Lipschitz map $\varphi\colon U \to U$ is a* ***deformation*** *supported in a compact set $K \subset U$ if:*

- *$\varphi(x) = x$ for all $x \in U \setminus K$;*
- *φ is smoothly isotopic to the identity (there exists a smooth map $H\colon [0,1] \times U \to U$ such that $H(0,\cdot) = \mathrm{Id}$ and $H(1,\cdot) = \varphi$).*

Definition 7.2 ((M,ε,δ)-minimal set)**.** *Let $E \subset \mathbb{R}^n$ be a closed set. Let $M \geq 1$, $\varepsilon > 0$, and $\delta > 0$. We say E is an (M,ε,δ)-**minimal set** (with respect to a boundary Γ) if, for any Lipschitz deformation φ supported in a ball $B_r(x)$ (with $x \in E \setminus \Gamma$ and $r < \delta$), the following energy comparison holds:*

$$\mathcal{H}^k(E \cap B_r(x)) \leq M\,\mathcal{H}^k(\varphi(E \cap B_r(x))) + \varepsilon\, r^k. \tag{7}$$

This inequality is the crux of the theory:

- If $M = 1$ and $\varepsilon = 0$, E is a **strict area minimizer** (cannot be improved by any deformation).
- If $M > 1$ or $\varepsilon > 0$, E is a **quasi-minimizer**. This allows for small physical imperfections or energy barriers, but as $r \to 0$, the εr^k term vanishes faster than the area, forcing the tangent behaviour to be minimal.

A crucial consequence of this stability is the existence of the density.

Theorem 7.1 (Monotonicity)**.** *If E is an area-minimizing set ($M = 1, \varepsilon = 0$) inside a ball $B_R(x)$, then the density function*

$$\Theta_E(x,r) = \frac{\mathcal{H}^k(E \cap B_r(x))}{\omega_k r^k}$$

is monotonically non-decreasing in r (where ω_k is the volume of the unit ball).

This ensures that the density $\Theta_E^k(x) := \lim_{r\to 0} \Theta_E(x,r)$ exists everywhere.

Just as we use derivatives to approximate smooth functions with planes, we use **tangent cones** to approximate minimal sets. We define the blow-up of E at x as the limit of a sequence of dilations:

$$C_x = \lim_{r\to 0} \frac{E - x}{r}$$

(convergence is in the local Hausdorff distance). If E is an (M,ε,δ)-minimal set, any such limit C_x is a **Minimal Cone**, a set invariant under dilation that minimizes area. The structure of a soap film at the microscopic level is entirely determined by which minimal cones are stable in $\mathbb{R}^3$.

While Almgren defined the framework, it was Jean Taylor who provided the classification of these cones in 3D, turning Plateau's experimental observations into rigorous theorems.

Theorem 7.2 (Taylor, 1976)**.** *Let E be a 2-dimensional (M, ε, δ)-minimal set in $\mathbb{R}^3$. Then, for any point $x \in E \setminus \Gamma$, the tangent cone C_x is congruent to exactly one of three types:*

1. ***The Plane*** *(Type $\mathbb{P}$):*
   - *Density:* $\Theta_E^2(x) = 1$.
   - *Geometry: A standard flat plane.*
   - *Occurrence: This corresponds to regular points (the vast majority of the surface).*
2. ***The Triple Junction*** *(Type $\mathbb{Y}$) (Fig 5, left):*
   - *Density:* $\Theta_E^2(x) = \frac{3}{2} = 1.5$.
   - *Geometry: Three half-planes meeting along a common line (the singular edge).*
   - *Angles: The half-planes meet at dihedral angles of* $120°$.
3. ***The Tetrahedral Point*** *(Type $\mathbb{T}$) (Fig 5, right):*
   - *Density:* $\Theta_E^2(x) = \frac{1}{\pi}\left(3\arccos\left(-\frac{1}{3}\right) + \sqrt{2}\right) \approx 1.82$.
   - *Geometry: The cone over the 1-skeleton of a regular tetrahedron. It consists of* ***six planar sectors*** *bounded by four rays meeting at the origin.*
   - *Angles: The four rays (singular edges) meet at the tetrahedral angle* $\arccos(-1/3) \approx 109.47°$. *The six planar faces meet along these rays at* $120°$.

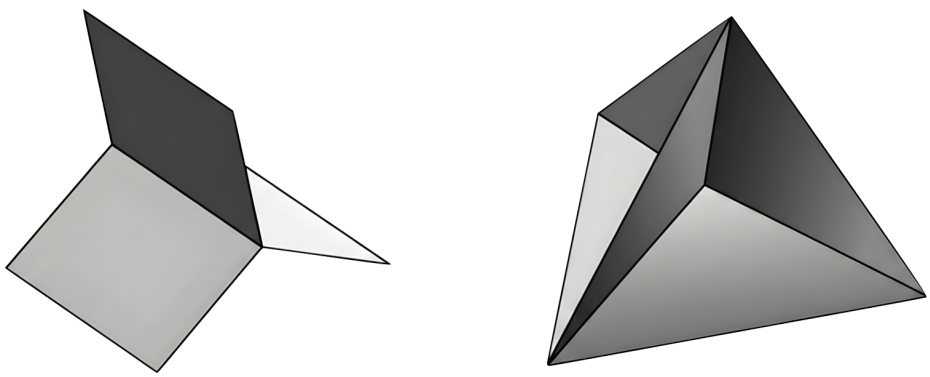

Figure 5: The only stable singularities in 3D soap films: Y-junction (3 sheets) on the left, and the T-junction (Tetrahedral point) on the right.

This theorem proves that the complex singularities observed by Plateau are the *only* possible stable local structures.

While Taylor solved the problem in $\mathbb{R}^3$, Fred Almgren devoted his life to solving the general regularity problem for minimal sets of dimension k in $\mathbb{R}^n$. He sought to prove that the singular set always has dimension at most $k-2$. He passed away in 1997 before seeing the final publication of his work. His magnum opus, simply known as *"Almgren's Big Regularity Paper"*, was published posthumously in 2000. It is a tour de force of geometric analysis, spanning 955 pages, and remains one of the deepest works in the field.

8 Predicting and Simulating Soap Films

We have travelled from the 18th-century calculus of variations to the 20th-century depth of Geometric Measure Theory. Now, we face the practical question: Given a boundary Γ, can we predict the shape of the resulting film?

Before turning to computers, three geometric principles allow us to guess the qualitative behaviour of a soap film.

1. **The Convex Hull Property:** A minimal surface S spanning a boundary Γ must lie entirely within the convex hull of Γ. *Mathematical justification:* The coordinate functions x^i of a minimal surface are harmonic on the surface ($\Delta_S x^i = 0$). By the Maximum Principle for harmonic functions, the maximum and minimum values of the coordinates must occur on the boundary. Thus, the surface cannot "bulge out" beyond the convex enclosure of the wire.

2. **Topological Phase Transitions:** Films are sensitive to the aspect ratio of the boundary. Consider the **Catenoid**: if we pull the two rings apart, there is a critical distance ($\approx 0.66\times$ circumference) where the neck becomes unstable. The film undergoes a topological surgery, collapsing the neck and separating into two disjoint disks (the Goldschmidt solution). This is a physical realization of a bifurcation in the energy landscape.

3. **The Steiner Skeleton:** For a disconnected boundary (e.g., four points), the film often resembles a "web" rather than a smooth sheet. The singular set (where the film branches) forms a **Steiner Tree**. While the film minimizes Area (2D), its cross-section effectively minimizes Length (1D). Therefore, to predict the singularity structure, one can solve the Steiner Tree problem for the boundary points: find the graph of minimal length connecting them. The soap film will effectively be the "fleshed out" version of this graph, respecting the 120° angles at the junctions.

While heuristics are useful, precise modeling requires numerical simulation. The challenge is significant: standard Finite Element Methods (FEM) struggle because the domain geometry is unknown (it is part of the solution) and the topology may change.

The gold standard in this field is the **Surface Evolver**, developed by Ken Brakke.

How it works:

1. **Discretization:** The surface is represented as a **simplicial complex** (a union of triangles).

2. **Energy Functional:** The program calculates the total energy E, which is usually surface tension ($\gamma \cdot \text{Area}$) but can include gravity, pressure, or crystalline anisotropy.

3. **Gradient Descent:** The vertices of the mesh are moved in the direction of the negative energy gradient:
$$\vec{x}_{new} = \vec{x}_{old} - \eta \nabla E(\vec{x}),$$
effectively simulating a mean curvature flow.

4. **Refinement and Surgery:** This is the crucial step. As the surface evolves, triangles may become extremely elongated or vanish. The software automatically performs:

   - **Refining:** Splitting large triangles;
   - **Vertex Averaging:** Removing small triangles;
   - **Topological Changes:** If a neck pinches off (radius $\to 0$), the mesh is locally disconnected to allow the separation of the surface.

This method allows mathematicians to "experiment" with boundaries that are too complex to build physically or too difficult to solve analytically.

The theory of minimal surfaces is not just an abstract delight; it has shaped our built environment and our understanding of materials.

The most famous application of soap film mechanics in architecture is the roof of the 1972 Munich Olympic Stadium. The architect Frei Otto pioneered the use of "tensile structures." In the pre-computer era, Otto could not solve the non-linear PDEs numerically. Instead, he used a method called **Form-Finding**:

- He built physical scale models using tulle fabric, wire, and soap films.
- By measuring the geometry of the soap film (which naturally found the equilibrium state), he determined the optimal shape for the tensile roof.
- Photogrammetry was used to upscale these coordinates to build the actual stadium.

The result is a structure where every square inch is in tension, minimizing material usage and maximizing structural efficiency (see Fig. 6

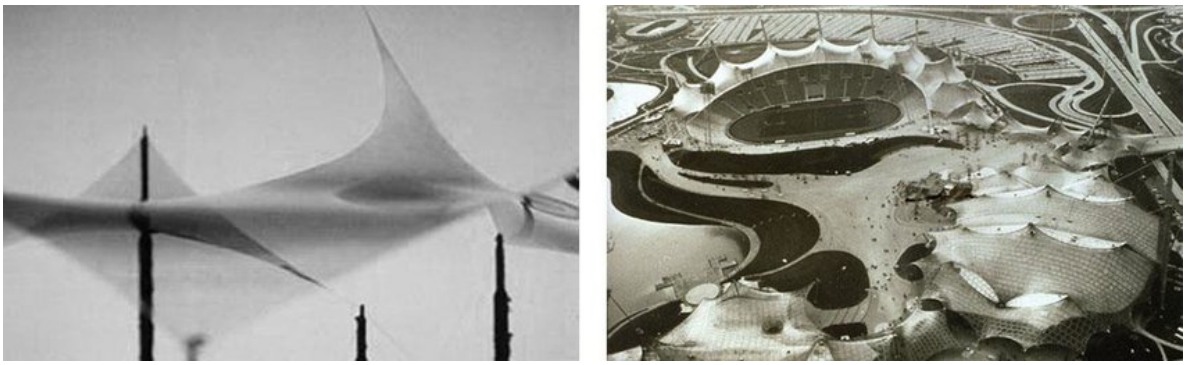

Figure 6: Munich Olympic Stadium.

In 1887, Lord Kelvin asked: *"What is the most efficient way to partition space into cells of equal volume with the least surface area?"* (i.e., what is the shape of a dry foam?). Kelvin proposed a structure of truncated octahedra. However, in 1993, Weaire and Phelan used numerical software (Surface Evolver!) to discover a structure with 0.3% less surface area.

- This **Weaire-Phelan structure** is physically realized in the National Aquatics Center (Water Cube) in Beijing (2008 Olympics).
- The building's façade is a slice through this optimal foam structure. The beams follow the Plateau borders, meeting at the required 109.47° angles, creating a structure that is both aesthetically organic and structurally rigid (see Fig 7).

Modern material science utilizes "metal foams"—structures created by injecting gas into molten metal.

- These materials inherit the geometry of soap froths (satisfying Plateau's laws during the liquid phase).
- They are incredibly lightweight but possess high stiffness and energy absorption (ideal for crash imperfections in cars).

Figure 7: Beijing National Aquatics Center.

- Understanding the geometry of the singularities (Plateau borders) is crucial, as mechanical failure often begins at the nodes where films meet.

When you dip a wire into soapy water, the liquid performs a computation that beats classical calculus. It solves a non-linear, free-boundary problem, minimizes area, automatically adjusts its topology (genus), and constructs a singularity set satisfying Jean Taylor's rigorous geometric laws.

Soap films are not just minimizing area; they are physical realizations of **Geometric Measure Theory**. Next time you wash your dishes, take a moment to look at the foam. You are looking at a solution to a problem that took Euler, Lagrange, Plateau, Douglas, Radó, Federer, Fleming, De Giorgi, Almgren, Taylor, and Allard over 250 years to understand.

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